Practice test

1. **Sample A: 47, 85, 32, 74, 58, 63, 91, 29, 76, 54, 68, 37**

   Sample B: 52, 79, 34, 61, 83, 45, 98, 27, 66, 72, 39, 54

   a) Apply the best dispersion measure on sample A and comment on your findings.

   b) Which sample is more homogeneous.

2. For the given data set bellow detect outlier/outliers

   428, 753, 92, 564, 571, 305, 689, 237, 990, 486, 621, 443

3. Following are the income and saving of 10 randomly selected student

Income	**150**	**210**	**180**	**190**	**160**	**250**	**220**	**175**	**205**	**170**
Saving	**20**	**28**	**27**	**25**	**21**	**32**	**30**	**25**	**24**	**22**

   a) Determine the strength and direction of the relationship between income and saving.

   b) Measure the Coefficient of skewness for income.

4. Following the distance from Dhaka to world most popular cities

   4827, 3619, 5073, 2145, 5294, 1432, 9761, 5582, 5314, 4950, 6406, 3298

   a) Measure the Coefficient of skewness.

   b) Determine variance from the data set.

   c) For the given data set detect outlier/outliers

5. Following are the temperature and sales of juice of 10 randomly selected day of a month,

Temperature	**30**	**29**	**26**	**31**	**35**	**34**	**37**	**39**	**36**	**32**
Sales	**210**	**200**	**180**	**215**	**255**	**260**	**295**	**310**	**280**	**220**

   a) Determine the strength and direction of the relationship between temperature and sales.

   b) Measure the Coefficient of skewness for sales.

c) Determine variance for temperature,

d) Determine outlier for sales,

6. The sales of units by two shops are given by-

Sample A: 870, 750, 320, 740, 860, 930, 1060, 802, 760, 785, 608, 1206

Sample B: 586, 685, 780, 610, 830, 450, 980, 270, 640, 720, 930, 720

a) Apply the best dispersion measure on sample A and comment on your findings.

b) Which sample is more consistent?

c) Determine outlier for both samples.